

$$f(x) = \sin(x^2)$$

$$\frac{df}{dx}(x) = 2x \cos(x^2) \longrightarrow \text{Manual differentiation (Exact, ground truth)}$$

Approximating derivatives

Finite difference formula

h is a small value

$$\frac{df}{dx}(x_i) \approx \frac{f(x_i + h) - f(x_i)}{h}$$

Forward difference scheme

$$\frac{df}{dx}(x_i) \approx \frac{f(x_i) - f(x_i - h)}{h}$$

Backward difference scheme

$$\frac{df}{dx}(x_i) \approx \frac{f(x_i + h) - f(x_i - h)}{2h}$$

Central difference scheme

$$f(x) = \sin(x^2) \quad \frac{df}{dx}(x) = 2x \cos(x^2)$$

$$\text{At } x_i = \pi/2$$

$$\frac{df}{dx}(x = \pi/2) = 2 \frac{\pi}{2} \cos\left(\frac{\pi^2}{4}\right) = -2.4542495411512917$$

Exact value

$$h = 0.001$$

Forward difference scheme

$$\frac{df}{dx}(x_i) \approx \frac{f(x_i + h) - f(x_i)}{h} = -2.4581093000713983$$

Backward difference scheme

$$\frac{df}{dx}(x_i) \approx \frac{f(x_i) - f(x_i - h)}{h} = -2.4503856304469718$$

Central difference scheme

$$\frac{df}{dx}(x_i) \approx \frac{f(x_i + h) - f(x_i - h)}{2h} = -2.454247465259185$$

Comparing the numbers above, backward and forward schemes have approximately the same accuracy. Usually central scheme is more accurate than forward and backward.

You can also compute the derivatives of a function using the automatic differentiation packages in Julia. The main ones are ForwardDiff and Zygote. These packages can give you the exact derivative (down to machine accuracy).

$$f(x) = \sin(x^2)$$

```
f(x)=sin(x^2);
```

```
ForwardDiff.derivative(f, π/2)
```

```
Zygote.gradient(f, π/2)[1]
```

```
julia> ForwardDiff.derivative(f, π/2)  
-2.4542495411512917
```

```
julia> Zygote.gradient(f, π/2)  
(-2.4542495411512917, )
```


Newton-Raphson

without having to manually differentiate!

```
f2(x)=4π*x-2/x^2
x=1.0
for i=1:10
    x-=f2(x)/ForwardDiff.derivative(f2,x)
    println(x)
end
```

$$f(x) = 4\pi x - 2/x^2$$

$$x_{i+1} = x_i - f(x_i)/f'(x_i)$$

$$h(x_1, x_2) = (x_1^2 + x_2 - 11)^2 + (x_1 + x_2^2 - 7)^2$$

```
himmelblau(x)=(x[1]^2+x[2]-11)^2+(x[1]+x[2]^2-7)^2
```

```
ForwardDiff.gradient(himmelblau, [1.0, 1.0])
```

2-element Vector{Float64}:

-46.0

-38.0

$$\frac{\partial h}{\partial x_1}(1,1)$$

$$\frac{\partial h}{\partial x_2}(1,1)$$

```
Zygote.gradient(himmelblau, [1.0, 1.0])[1]
```

2-element Vector{Float64}:

-46.0

-38.0

```
function NN(x,W,b)
    sin(W*x+b)
end
```

$$NN(x, W, b) = \sin(Wx + b)$$

```
ForwardDiff.derivative((x)->NN(x,2,3),1.0)
```

```
ForwardDiff.derivative((W)->NN(1,W,3),2.0)
```

```
ForwardDiff.derivative((b)->NN(1,2,b),3.0)
```

```
Zygote.gradient(NN,1,2,3)
```

(0.5673243709264525, 0.28366218546322625, 0.28366218546322625)

$$\frac{\partial NN}{\partial x}(x = 1, W = 2, b = 3)$$

$$\frac{\partial NN}{\partial W}(x = 1, W = 2, b = 3)$$

$$\frac{\partial NN}{\partial b}(x = 1, W = 2, b = 3)$$

$$\frac{\partial NN}{\partial x}(x = 1, W = 2, b = 3)$$

$$\frac{\partial NN}{\partial W}(x = 1, W = 2, b = 3)$$

$$\frac{\partial NN}{\partial b}(x = 1, W = 2, b = 3)$$

Vector functions of vectors!

`x=[1, 2, 4, 7, 11]`

5-element Vector{Int64}:

1
2
4
7
11

`x[2:end]`

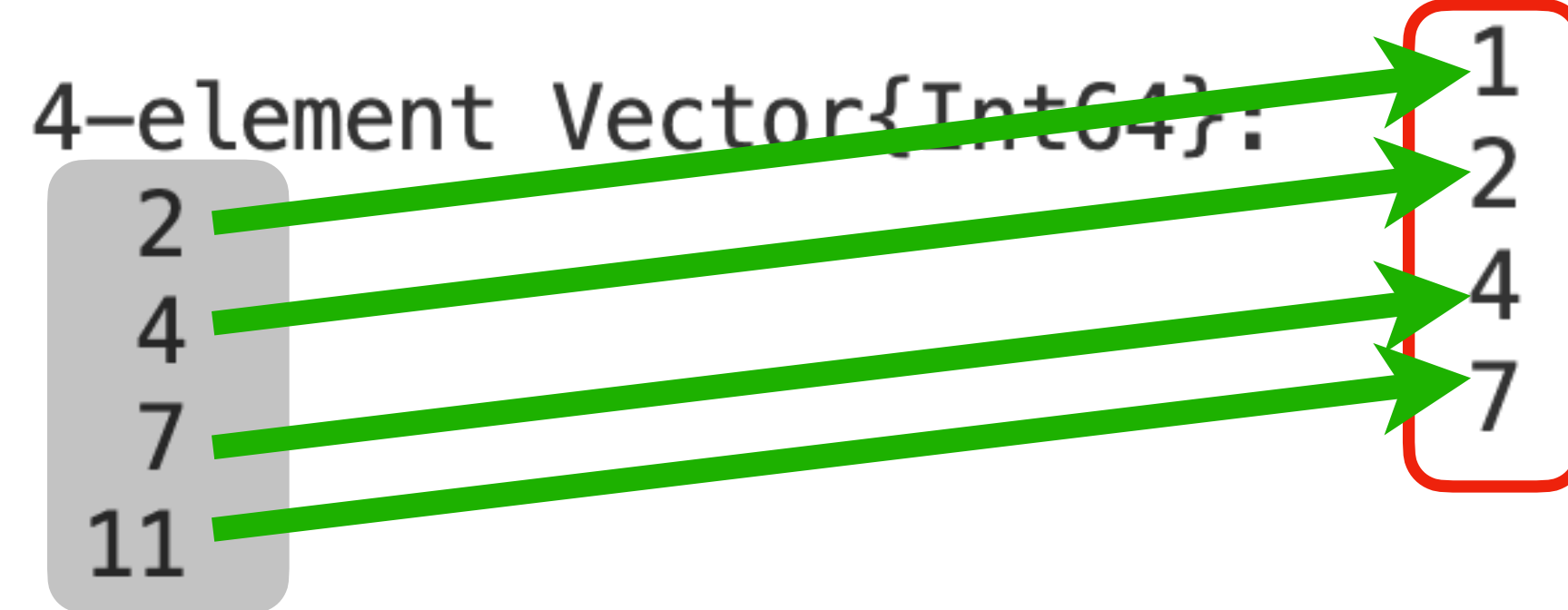
4-element Vector{Int64}:

2
4
7
11

`x[1:end-1]`

4-element Vector{Int64}:

1
2
4
7



`x[2:end]-x[1:end-1]`

4-element Vector{Int64}:

1
2
3
4

$$f_i(\vec{x}) = \begin{cases} x_i - 1 & i = 1 \\ x_i - x_{i-1}^2 & i = 2..N-1 \\ x_i - x_{i-1} & i = N \end{cases}$$

For $N = 5$

If $x_1 = x_2 = x_3 = x_4 = x_5 = 2$

$$\begin{Bmatrix} f_1(x_1, x_2, x_3, x_4, x_5) \\ f_2(x_1, x_2, x_3, x_4, x_5) \\ f_3(x_1, x_2, x_3, x_4, x_5) \\ f_4(x_1, x_2, x_3, x_4, x_5) \\ f_5(x_1, x_2, x_3, x_4, x_5) \end{Bmatrix} = \begin{Bmatrix} x_1 - 1 \\ x_2 - x_1^2 \\ x_3 - x_2^2 \\ x_4 - x_3^2 \\ x_5 - x_4 \end{Bmatrix} = \begin{Bmatrix} 1 \\ -2 \\ -2 \\ -2 \\ 0 \end{Bmatrix}$$

```
fvec2(x)=[x[1]-1;  
x[2:end-1].-x[1:end-2].^2;  
x[end]-x[end-1]]
```

```
N=5  
x=2*ones(N)  
fvec2(x)
```

$$\begin{Bmatrix} f_1(x_1, x_2, x_3, x_4, x_5) \\ f_2(x_1, x_2, x_3, x_4, x_5) \\ f_3(x_1, x_2, x_3, x_4, x_5) \\ f_4(x_1, x_2, x_3, x_4, x_5) \\ f_5(x_1, x_2, x_3, x_4, x_5) \end{Bmatrix} = \begin{Bmatrix} x_1 - 1 \\ x_2 - x_1^2 \\ x_3 - x_2^2 \\ x_4 - x_3^2 \\ x_5 - x_4 \end{Bmatrix}$$

ForwardDiff.jacobian(fvec2,x)

5x5 Matrix{Float64}:

1.0	0.0	0.0	0.0	0.0
-4.0	1.0	0.0	0.0	0.0
0.0	-4.0	1.0	0.0	0.0
0.0	0.0	-4.0	1.0	0.0
0.0	0.0	0.0	-1.0	1.0

$$\partial \vec{f} / \partial \vec{x} = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} & \frac{\partial f_1}{\partial x_3} & \frac{\partial f_1}{\partial x_4} & \frac{\partial f_1}{\partial x_5} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} & \frac{\partial f_2}{\partial x_3} & \frac{\partial f_2}{\partial x_4} & \frac{\partial f_2}{\partial x_5} \\ \frac{\partial f_3}{\partial x_1} & \frac{\partial f_3}{\partial x_2} & \frac{\partial f_3}{\partial x_3} & \frac{\partial f_3}{\partial x_4} & \frac{\partial f_3}{\partial x_5} \\ \frac{\partial f_4}{\partial x_1} & \frac{\partial f_4}{\partial x_2} & \frac{\partial f_4}{\partial x_3} & \frac{\partial f_4}{\partial x_4} & \frac{\partial f_4}{\partial x_5} \\ \frac{\partial f_5}{\partial x_1} & \frac{\partial f_5}{\partial x_2} & \frac{\partial f_5}{\partial x_3} & \frac{\partial f_5}{\partial x_4} & \frac{\partial f_5}{\partial x_5} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ -2x_1 & 1 & 0 & 0 & 0 \\ 0 & -2x_2 & 1 & 0 & 0 \\ 0 & 0 & -2x_3 & 1 & 0 \\ 0 & 0 & 0 & -1 & 1 \end{bmatrix}$$

```
using ForwardDiff
```

```
VectorFunc(x)=[x[1]^2-x[2]+1,  
               3*cos(x[1])-x[2]]
```

```
x=[1,1]
```

```
for i=1:5
```

```
    F=VectorFunc(x)
```

```
    J=ForwardDiff.jacobian(VectorFunc,x)
```

```
    delta=-J\F
```

```
    x+=delta
```

```
    println(x)
```

```
end
```

$$f_1(\vec{x}) = x_1^2 - x_2 + 1 = 0$$

$$f_2(\vec{x}) = 3 \cos(x_1) - x_2 = 0$$

$$\begin{bmatrix} \partial f_1 / \partial x_1 & \partial f_1 / \partial x_2 \\ \partial f_2 / \partial x_1 & \partial f_2 / \partial x_2 \end{bmatrix} \begin{Bmatrix} x_1^{(i+1)} - x_1^{(i)} \\ x_2^{(i+1)} - x_2^{(i)} \end{Bmatrix} = - \begin{Bmatrix} f_1(x_1^{(i)}, x_2^{(i)}) \\ f_2(x_1^{(i)}, x_2^{(i)}) \end{Bmatrix}$$